IoT-Enabled Smart Bus Route Monitoring and Revenue Management System



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PP1 FEEDBACK MATRIX

Supervisor Feedback	How we addressed them
Clarify integrated system architecture	Integrated all individual components
Stick to given presentation time	Practiced the presentation multiple times

PROBLEM STATEMENT

CORE PROBLEM

Low financial sustainability causes unreliable public bus operations during night time

OTHER CHALLENGES

- Buses deviate from their permitted road during operation hours
- Bus owners only have to pay a significantly low amout annually to maintain their permits
- Manual schedule creation consume valuable time daily
- Third party involvement (Grama Niladari) causes curruption
- Unsafe driving behavior causes passenger discomfort

1

Transport Authority

Monitoring, revenue collection and dynamic scheduling

2

Bus Owners

Revenue, incentives, and operational feasibility

3

Drivers

Trip tracking, real-time alerts, and violation logs

4

Passengers / Public

Reliable transport availability & service guarantees

RESEARCH GAP & OBJECTIVES

EXISTING SYSTEMS FOCUS ON

Individual aspects of transportation such as basic vehicle tracking using mobile GPS , ETA predictions, and passenger information systems.

THE MISSING GAP

Lack of an integrated system that uses verified real world bus operation to dynamically calculate permit fees and financially incentivize reliable night time service

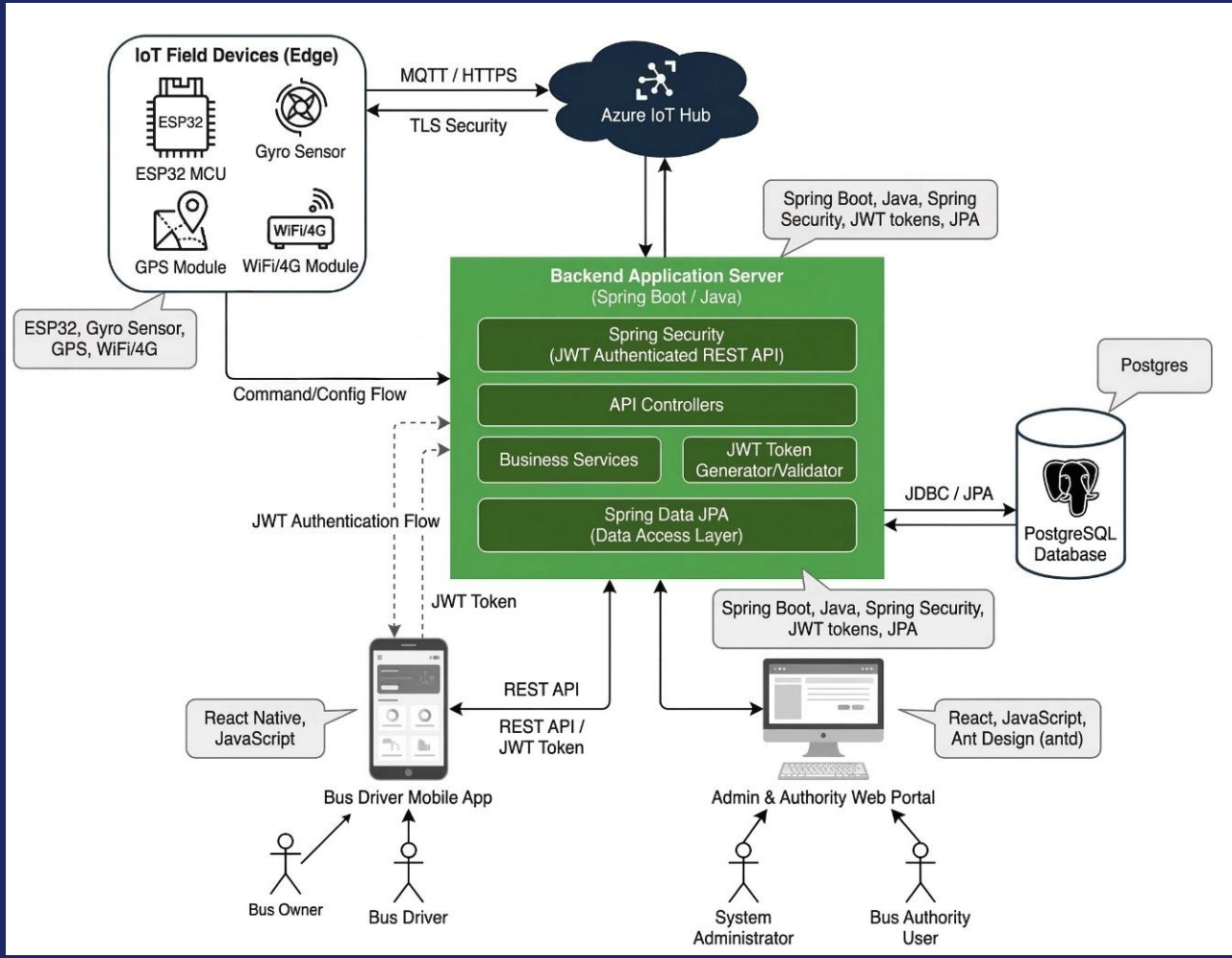
RO1 Develop an event-driven IoT device with GPS and Gyro sensors for bus operational monitoring.

RO2 Build a web application that allows authorities to create schedules, track bus operations and monitor violations .

RO3 Develop a distance based permit fee and commision calculation model to improve reliability of night time operations.

RO4 Develop a mobile application to support bus drivers and owners to see their work and get notified.

SOLUTION ARCHITECTURE



01

Policy-Driven Dynamic Scheduling

Peak / Normal / Off-Peak / Night + disruption handling.

02

Event-Driven IoT Monitoring

Trip, halt, deviation, and movement events.

03

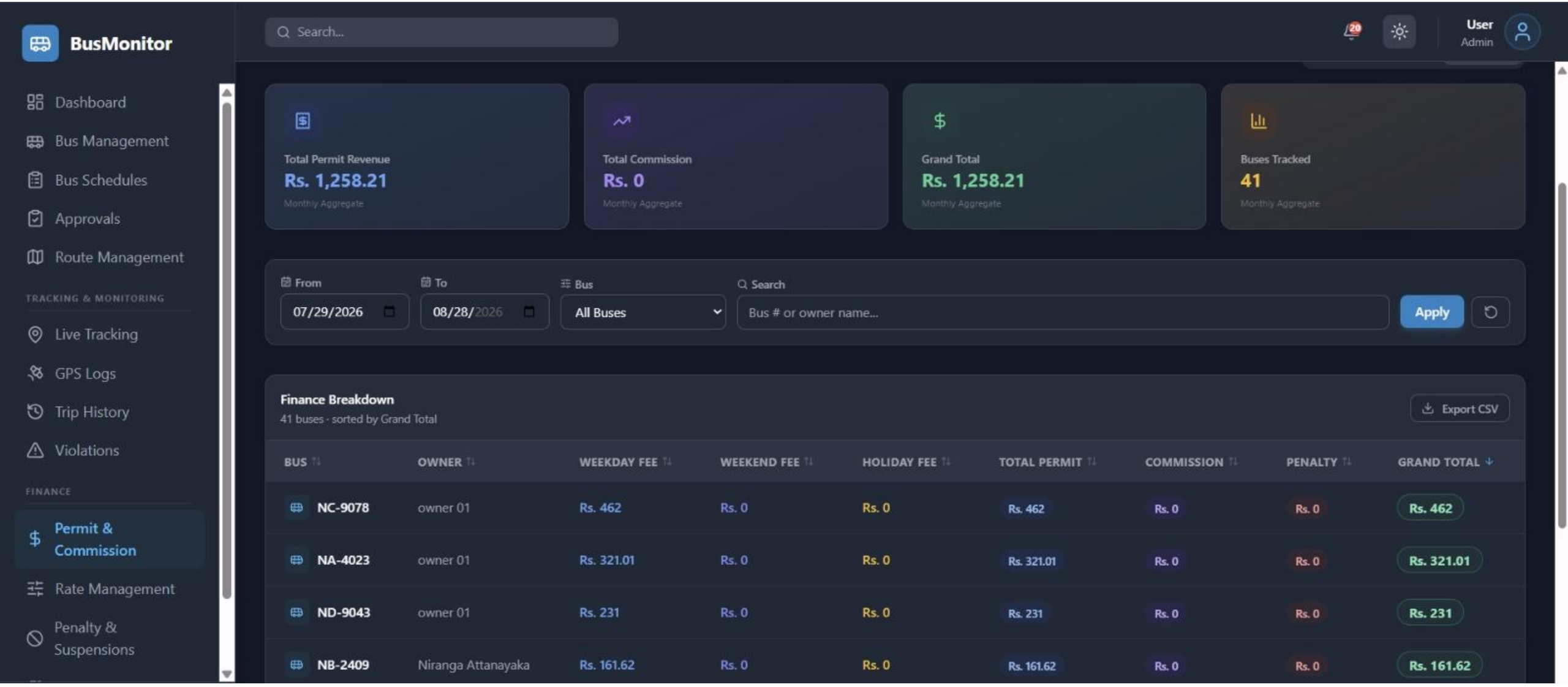
Driver Behaviour Monitoring

Speeding, harsh acceleration/braking, and route violations.

04

Distance-Based Revenue Engine

Actual operational distance + policy-based calculation.


BusMonitor
Search...
User
Admin
Dashboard
Bus Management
Bus Schedules
Approvals
Route Management
TRACKING & MONITORING
Live Tracking
GPS Logs
Trip History
Violations
FINANCE
Permit & Commission
Rate Management
Penalty & Suspensions
Total Permit Revenue
Rs. 1,258.21
Monthly Aggregate
Total Commission
Rs. 0
Monthly Aggregate
Grand Total
Rs. 1,258.21
Monthly Aggregate
Buses Tracked
41
Monthly Aggregate
From
07/29/2026
To
08/28/2026
Bus
All Buses
Search
Bus # or owner name...
Apply
Finance Breakdown
41 buses · sorted by Grand Total
Export CSV
BUS
OWNER
WEEKDAY FEE
WEEKEND FEE
HOLIDAY FEE
TOTAL PERMIT
COMMISSION
PENALTY
GRAND TOTAL
NC-9078 owner 01 Rs. 462 Rs. 0 Rs. 0 Rs. 462 Rs. 0 Rs. 0 Rs. 462
NA-4023 owner 01 Rs. 321.01 Rs. 0 Rs. 0 Rs. 321.01 Rs. 0 Rs. 0 Rs. 321.01
ND-9043 owner 01 Rs. 231 Rs. 0 Rs. 0 Rs. 231 Rs. 0 Rs. 0 Rs. 231
NB-2409 Niranga Attanayaka Rs. 161.62 Rs. 0 Rs. 0 Rs. 161.62 Rs. 0 Rs. 0 Rs. 161.62

BEYOND THE CODE

Commercialization, AI Policy & Ethics

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Commercial Value

- Target customers: transport authorities, bus operator associations
- Revenue model: SaaS platform subscription, per-bus monthly monitoring fee, and hardware setup / leasing fee.

AI

AI Policy Disclosure

- Assisted tools: Claude AI and chat gpt was used strictly for documentation assistance, language refinement and code debugging.
- Human ownership: all research decisions, system designs, algorithms, test executions, and results were performed and verified by the student team.

E

Ethics & Security

- Driver privacy: location monitoring strictly restricted to active service shifts.
- Data protection: encrypted transmission of operational & revenue logs.
- Fairness: transparent violation logging and audit trails for penalty disputes.

CONCLUSION

Transforming Bus Operations Through Data-Driven Intelligence

FINANCIAL

Enables distance-based revenue and policy incentives.

OPERATIONAL

Enforces route compliance and dynamic scheduling during disruptions.

MANAGEMENT

Provides verified, real-time evidence for authority decision-making.

“Our goal is to make public bus operations more transparent, financially sustainable, and responsive to transportation demand.”

🎬 LIVE DEMO & TECHNICAL DEFENCE

Demonstrating: IoT Telemetry → Route Violation Detection → Dynamic Schedule Re-adjustment → Distance Revenue Calculation